

Combinatorial Identity

A Session in Combinatorial Structures

ISI UGB Mastery

Combinatorics · Polynomials · Number Theory

What This Class Is About

- A single identity that connects
 - binomial sums,
 - number theory,
 - integrals of polynomials,
 - divisibility of binomial coefficients.
- The identity is elementary. The applications are not.
- Goal for the class: recognise the shape of a sum that yields to this identity, and learn the standard manoeuvres that follow from it.

The Core Identity

Identity 1: Committee–Subcommittee

For non-negative integers n, k, m with $n \geq k \geq m$,

$$\binom{n}{k} \binom{k}{m} = \binom{n}{m} \binom{n-m}{k-m}.$$

- Left side: choose a committee of k , then an executive of m inside it.
- Right side: choose the executive of m first, then fill the remaining $k - m$ committee seats from the $n - m$ non-executives.
- Both procedures count the same object: a pair (committee, executive).

Proof 1: The Combinatorial Argument

Double counting

Count the same set of configurations in two different orders.

A configuration is a pair (C, E) where $C \subseteq \{1, \dots, n\}$, $|C| = k$, and $E \subseteq C$, $|E| = m$.

Order 1. Pick C first: $\binom{n}{k}$ ways. Then pick $E \subseteq C$: $\binom{k}{m}$ ways.

Order 2. Pick E first: $\binom{n}{m}$ ways. Then pick the remaining $k - m$ members of C from the $n - m$ elements outside E : $\binom{n-m}{k-m}$ ways.

Both orders count the same set of pairs, so the two products agree.

Proof 2: The Algebraic Argument

Expand using factorials:

$$\binom{n}{k} \binom{k}{m} = \frac{n!}{k!(n-k)!} \cdot \frac{k!}{m!(k-m)!} = \frac{n!}{m!(k-m)!(n-k)!}.$$

Now regroup the factorials by inserting $(n-m)!/(n-m)!$:

$$\frac{n!}{m!(n-m)!} \cdot \frac{(n-m)!}{(k-m)!(n-k)!} = \binom{n}{m} \binom{n-m}{k-m}.$$

Why two proofs

The combinatorial proof shows what the identity *means*. The algebraic proof shows that the identity is really a statement about regrouping a single factorial expression — which is why it can appear in so many disguises.

The Core Manoeuvre

The move

In the expression $\binom{n}{k}\binom{k}{m}$, the index k appears in *two* places. After applying the identity,

$$\binom{n}{k}\binom{k}{m} = \binom{n}{m}\binom{n-m}{k-m},$$

the index k appears in only *one* place. The factor $\binom{n}{m}$ is constant in k and can be pulled out of any sum over k .

This is the reason the identity is so effective for sums: it **free**s a summation index. Every application that follows is a variation on this one move.

Application 1 — Problem

Problem

Evaluate

$$S = \sum_{k=m}^n \binom{n}{k} \binom{k}{m}.$$

Think: the index k sits inside two binomial coefficients. Before we can sum, we must isolate k to one place.

Application 1 — Solution

Solution

Apply the identity to pull $\binom{n}{m}$ out of the sum:

$$S = \sum_{k=m}^n \binom{n}{m} \binom{n-m}{k-m} = \binom{n}{m} \sum_{k=m}^n \binom{n-m}{k-m}.$$

Substitute $j = k - m$, so the inner sum runs over $j = 0, \dots, n - m$:

$$\sum_{j=0}^{n-m} \binom{n-m}{j} = 2^{n-m}.$$

Therefore

$$S = \binom{n}{m} 2^{n-m}.$$

Application 2 — Problem

Binomial inversion / orthogonality

Prove the orthogonality relation

$$\sum_{k=m}^n (-1)^{k-m} \binom{n}{k} \binom{k}{m} = \delta_{n,m},$$

where $\delta_{n,m}$ is 1 if $n = m$ and 0 otherwise.

The identity on its own is algebraic; the signs $(-1)^{k-m}$ will turn a sum into a value of $(1 - 1)^{n-m}$ after the standard move.

Application 2 — Solution

Solution

Apply the identity and pull out the constant factor:

$$\sum_{k=m}^n (-1)^{k-m} \binom{n}{m} \binom{n-m}{k-m} = \binom{n}{m} \sum_{k=m}^n (-1)^{k-m} \binom{n-m}{k-m}.$$

With $j = k - m$, the inner sum becomes $\sum_{j=0}^{n-m} (-1)^j \binom{n-m}{j} = (1 - 1)^{n-m}$. This equals 0 for $n > m$ and 1 for $n = m$. Since $\binom{m}{m} = 1$, the full sum equals $\delta_{n,m}$.

Why this matters

This is the statement that the matrices $\left[\binom{n}{k}\right]$ and $\left[(-1)^{n-k} \binom{n}{k}\right]$ are inverses. That is the Möbius inversion of the Boolean lattice.

Application 3 — Problem

Nested binomial to multinomial

Starting from $(x + y + z)^n = (x + (y + z))^n$, show that the coefficient of $x^{n-k}y^mz^{k-m}$ matches the multinomial coefficient

$$\binom{n}{n-k, m, k-m} = \frac{n!}{(n-k)! m! (k-m)!}.$$

Application 3 — Solution

Solution

Expand in two stages:

$$(x + (y + z))^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} (y + z)^k = \sum_{k=0}^n \binom{n}{k} x^{n-k} \sum_{m=0}^k \binom{k}{m} y^m z^{k-m}.$$

The coefficient of $x^{n-k} y^m z^{k-m}$ is $\binom{n}{k} \binom{k}{m}$. By the identity this equals

$$\binom{n}{k} \binom{k}{m} = \frac{n!}{m! (k-m)! (n-k)!} = \binom{n}{n-k, m, k-m}.$$

So the nested binomial structure is exactly the multinomial structure, with no new theorem required.

Polynomial Weights from Small m

Set $m = 1$ and $m = 2$ in the identity.

Case $m = 1$ (committee with a president):

$$\binom{n}{k} \binom{k}{1} = \binom{n}{1} \binom{n-1}{k-1} \implies k \binom{n}{k} = n \binom{n-1}{k-1}.$$

Case $m = 2$ (committee with a two-person executive):

$$\binom{n}{k} \binom{k}{2} = \binom{n}{2} \binom{n-2}{k-2} \implies k(k-1) \binom{n}{k} = n(n-1) \binom{n-2}{k-2}.$$

What these absorb

These identities convert a polynomial-weighted sum $\sum p(k) \binom{n}{k}$ into a plain binomial sum. The trick is to write $p(k)$ as a combination of falling factorials $k, k(k-1), k(k-1)(k-2), \dots$

Application 4 — Problem

Problem

Evaluate

$$\sum_{k=0}^n k \binom{n}{k}.$$

Standard approaches: differentiate $(1+x)^n$ at $x=1$. We will use the committee-subcommittee identity instead, which needs no calculus.

Application 4 — Solution

Solution

Use $k\binom{n}{k} = n\binom{n-1}{k-1}$, valid for $k \geq 1$ (and the $k = 0$ term is zero). Then

$$\sum_{k=0}^n k\binom{n}{k} = \sum_{k=1}^n n\binom{n-1}{k-1} = n \sum_{j=0}^{n-1} \binom{n-1}{j} = n2^{n-1}.$$

Therefore

$$\sum_{k=0}^n k\binom{n}{k} = n2^{n-1}.$$

Application 5 — Problem

Problem

Evaluate

$$\sum_{k=0}^n k^2 \binom{n}{k}.$$

The square k^2 does not match either of our absorption identities directly. We must rewrite it in terms of falling factorials.

Application 5 — Solution

Solution

Write $k^2 = k(k-1) + k$. Then

$$\sum_{k=0}^n k^2 \binom{n}{k} = \sum_{k=2}^n k(k-1) \binom{n}{k} + \sum_{k=1}^n k \binom{n}{k}.$$

Apply the $m=2$ and $m=1$ absorptions:

$$n(n-1) \sum_{k=2}^n \binom{n-2}{k-2} + n \sum_{k=1}^n \binom{n-1}{k-1} = n(n-1)2^{n-2} + n2^{n-1}.$$

Factor out $n2^{n-2}$, using $2^{n-1} = 2 \cdot 2^{n-2}$, to get $n2^{n-2}((n-1) + 2) = n(n+1)2^{n-2}$. Therefore

$$\boxed{\sum_{k=0}^n k^2 \binom{n}{k} = n(n+1)2^{n-2}.}$$

The Absorption Identity

Starting from $k\binom{n}{k} = n\binom{n-1}{k-1}$ and shifting $(n, k) \mapsto (n+1, k+1)$:

$$(k+1)\binom{n+1}{k+1} = (n+1)\binom{n}{k} \implies \frac{1}{k+1}\binom{n}{k} = \frac{1}{n+1}\binom{n+1}{k+1}.$$

The point

A factor of $\frac{1}{k+1}$ inside a binomial sum is absorbed into the binomial itself, with a shift of parameters. This replaces the standard calculus trick of integrating $(1+x)^n$ and evaluating at a specific point.

Application 6 — Problem

Harmonic-type sum

Evaluate

$$\sum_{k=0}^n \frac{1}{k+1} \binom{n}{k}.$$

The $\frac{1}{k+1}$ factor is the signature of integration — but we will handle it algebraically with absorption.

Application 6 — Solution

Solution

Apply the absorption identity term by term:

$$\sum_{k=0}^n \frac{1}{k+1} \binom{n}{k} = \sum_{k=0}^n \frac{1}{n+1} \binom{n+1}{k+1} = \frac{1}{n+1} \sum_{j=1}^{n+1} \binom{n+1}{j}.$$

The inner sum is $2^{n+1} - \binom{n+1}{0} = 2^{n+1} - 1$. Therefore

$$\boxed{\sum_{k=0}^n \frac{1}{k+1} \binom{n}{k} = \frac{2^{n+1} - 1}{n+1}.}$$

Cross-check

The same sum equals $\int_0^1 (1+x)^n dx$. The agreement is not a coincidence: the absorption identity is the discrete shadow of the integration-by-one trick.

Application 7 — Problem

Generating polynomial

Evaluate, in closed form,

$$P(x) = \sum_{k=m}^n \binom{n}{k} \binom{k}{m} x^k.$$

The sum has the same structure as Application 1, now weighted by x^k . We expect a clean product-of-factors answer.

Application 7 — Solution

Solution

Apply the identity and pull out the constant factor:

$$P(x) = \binom{n}{m} \sum_{k=m}^n \binom{n-m}{k-m} x^k = \binom{n}{m} x^m \sum_{k=m}^n \binom{n-m}{k-m} x^{k-m}.$$

With $j = k - m$, the inner sum is $\sum_{j=0}^{n-m} \binom{n-m}{j} x^j = (1+x)^{n-m}$. Therefore

$$P(x) = \binom{n}{m} x^m (1+x)^{n-m}.$$

Setting $x = 1$ recovers Application 1. The closed form is the whole polynomial, not just its value at a point.

Application 8 — Problem

Alternating rational sum

Evaluate

$$\sum_{k=m}^n \frac{(-1)^{k-m}}{k+1} \binom{n}{k} \binom{k}{m}.$$

Two features: the sign $(-1)^{k-m}$ and the factor $\frac{1}{k+1}$. We will combine the generating polynomial of Application 7 with integration on $[0, 1]$.

Application 8 — Solution

Solution

From Application 7, $\sum_{k=m}^n \binom{n}{k} \binom{k}{m} x^k = \binom{n}{m} x^m (1+x)^{n-m}$. Replace x by $-x$ and rearrange signs:

$$\sum_{k=m}^n (-1)^{k-m} \binom{n}{k} \binom{k}{m} x^k = \binom{n}{m} x^m (1-x)^{n-m}.$$

Integrate both sides from 0 to 1:

$$\sum_{k=m}^n \frac{(-1)^{k-m}}{k+1} \binom{n}{k} \binom{k}{m} = \binom{n}{m} \int_0^1 x^m (1-x)^{n-m} dx = \binom{n}{m} \cdot \frac{m! (n-m)!}{(n+1)!}.$$

The second equality uses the Beta function $B(m+1, n-m+1)$. The factorials collapse:

$$\frac{n!}{m! (n-m)!} \cdot \frac{m! (n-m)!}{(n+1)!} = \frac{1}{n+1}.$$

So the sum equals $\boxed{\frac{1}{n+1}}$, independent of m .

Why Application 8 Is Remarkable

The answer $\frac{1}{n+1}$ does not depend on m .

The principle

A sum with two parameters n and m , several binomial coefficients, an alternating sign, and a rational weight — and the answer is a simple function of n alone.

This is the hallmark of a genuine identity: the mess on the left and the simplicity on the right are both true statements about the same quantity, and the identity is what makes the simplification possible.

A direct attack by induction on m is painful. The generating polynomial $\binom{n}{m}x^m(1-x)^{n-m}$ and the Beta integral together make the answer transparent.

Bridge to Number Theory

Everything so far has been about sums and integrals. We now turn to divisibility.

The identity

$$\binom{n}{k} \binom{k}{m} = \binom{n}{m} \binom{n-m}{k-m}$$

is an equation between positive integers. It therefore forces divisibility relations: the left side is a multiple of $\binom{n}{k}$, and so is the right side. Reading the identity as a divisibility relation gives us two theorems.

Theorem 1 — Statement

Pairwise common factors

For any integer $n > 2$ and any distinct k, m with $0 < k, m < n$,

$$\gcd\left(\binom{n}{k}, \binom{n}{m}\right) > 1.$$

In words: no two interior entries of a row of Pascal's triangle (beyond $n = 2$) are coprime.

Before the proof, note how striking this is: the $\binom{n}{k}$ are built from a single factorial, yet their internal structure is constrained enough that they always share a factor.

Theorem 1 — Proof

Proof

Assume $k > m$. The identity gives $\binom{n}{k}\binom{k}{m} = \binom{n}{m}\binom{n-m}{k-m}$, so $\binom{n}{k}$ divides the right side.

Suppose for contradiction that $\gcd(\binom{n}{k}, \binom{n}{m}) = 1$. Then by Euclid's lemma, $\binom{n}{k}$ must divide $\binom{n-m}{k-m}$.

But $0 < m$ and $0 < k < n$, so $\binom{n-m}{k-m} < \binom{n}{k}$ — the range we choose from has shrunk by $m > 0$ elements. A positive integer cannot be divided by a strictly larger one.

Contradiction. Hence $\gcd(\binom{n}{k}, \binom{n}{m}) > 1$.

Theorem 2 — Statement

Divisibility lemma

For positive integers n, k with $k \leq n$,

$$\frac{n}{\gcd(n, k)} \text{ divides } \binom{n}{k}.$$

This is a refined statement of the fact that $\binom{n}{k}$ is an integer, weighted by how much of n the index k shares.

Theorem 2 — Proof

Proof

From $m = 1$, $k\binom{n}{k} = n\binom{n-1}{k-1}$. Let $d = \gcd(n, k)$ and write $n = dn'$, $k = dk'$ with $\gcd(n', k') = 1$. The identity becomes

$$dk'\binom{n}{k} = dn'\binom{n-1}{k-1} \implies k'\binom{n}{k} = n'\binom{n-1}{k-1}.$$

Thus n' divides $k'\binom{n}{k}$. Since $\gcd(n', k') = 1$, Euclid's lemma forces $n' \mid \binom{n}{k}$.

But $n' = n / \gcd(n, k)$, so $\frac{n}{\gcd(n, k)}$ divides $\binom{n}{k}$.

Corollary: The Freshman's Dream

Application of Theorem 2

For a prime p and $0 < k < p^a$, the binomial coefficient $\binom{p^a}{k}$ is divisible by p .

Proof. By Theorem 2, $\frac{p^a}{\gcd(p^a, k)}$ divides $\binom{p^a}{k}$. Since $0 < k < p^a$, we have $\gcd(p^a, k) = p^b$ for some $b < a$. Hence $\frac{p^a}{p^b} = p^{a-b}$ divides $\binom{p^a}{k}$, and $a - b \geq 1$.

Consequence. In characteristic p ,

$$(x + y)^{p^a} \equiv x^{p^a} + y^{p^a} \pmod{p}.$$

This is the 'Frobenius endomorphism', and it underpins much of algebraic geometry in positive characteristic. All of it rests on the committee-subcommittee identity with $m = 1$.

Variant 1: Disjoint Committees

Identity 2: Disjoint Committees

$$\binom{n}{k} \binom{n-k}{m} = \binom{n}{m} \binom{n-m}{k} = \frac{n!}{k! m! (n-k-m)!}.$$

Story. Partition n people into three groups: committee A of size k , committee B of size m , and the rest. The identity says: it does not matter which committee we pick first.

Why it is useful. It evaluates double sums that look intractable. For example,

$$\sum_{k+m \leq n} \binom{n}{k} \binom{n-k}{m} x^k y^m = (1+x+y)^n,$$

by the multinomial theorem. The identity is the combinatorial proof of the multinomial theorem for three terms.

Variant 1 — Worked Example, Problem

Problem

Evaluate $\sum_{k+m \leq n} \binom{n}{k} \binom{n-k}{m} 2^k 3^m$.

The sum runs over pairs (k, m) with $k + m \leq n$. With two running indices and two nested binomials, the right move is to recognise a multinomial.

Variant 1 — Worked Example, Solution

Solution

By the disjoint-committees identity,

$$\binom{n}{k} \binom{n-k}{m} = \frac{n!}{k! m! (n-k-m)!}.$$

The sum is therefore a trinomial expansion. Let $\ell = n - k - m$:

$$\sum_{k+m+\ell=n} \frac{n!}{k! m! \ell!} 2^k 3^m 1^\ell = (2 + 3 + 1)^n = 6^n.$$

The lesson

A double sum with two running indices inside two nested binomials becomes a single multinomial expansion once we recognise the structure.

Variant 2: Iterated Subcommittees

Identity 3: Chains of Subsets

For $n \geq a_1 \geq a_2 \geq \cdots \geq a_r \geq 0$,

$$\binom{n}{a_1} \binom{a_1}{a_2} \cdots \binom{a_{r-1}}{a_r} = \binom{n}{a_r} \binom{n - a_r}{a_{r-1} - a_r} \cdots \binom{n - a_2}{a_1 - a_2}.$$

The left side counts nested chains $A_r \subseteq A_{r-1} \subseteq \cdots \subseteq A_1 \subseteq \{1, \dots, n\}$ with prescribed sizes.

The right side counts the same chains by independently assigning each element to a “shell”:

- A_r ,
- $A_{r-1} \setminus A_r$,
- $A_{r-2} \setminus A_{r-1}$,
- $\dots$,
- outside A_1 .

Each element is placed independently into one of $r + 1$ possible shells.

Variant 2 — Counting Chains, Problem

Problem

Let $n \geq 0$ and $r \geq 0$. Count the number of chains

$$A_r \subseteq A_{r-1} \subseteq \cdots \subseteq A_1 \subseteq \{1, \dots, n\}$$

where the sizes $a_i = |A_i|$ are not fixed in advance.

The sizes are free, so summing over all $(a_1, \dots, a_r)$ with Σ the identity above gives many nested binomials. But Identity 3 suggests a much simpler route: look at where each element can go.

Variant 2 — Counting Chains, Solution

Solution

Each element of $\{1, \dots, n\}$ independently lands in exactly one of $r + 1$ regions:

- A_r ,
- $A_{r-1} \setminus A_r, A_{r-2} \setminus A_{r-1}, \dots, A_1 \setminus A_2$,
- outside A_1 .

Since the n elements are placed independently, the total number of chains is

$$(r + 1)^n.$$

Putting the Three Identities Together

Identity	What it does
$\frac{\binom{n}{k} \binom{k}{m}}{\binom{n}{m} \binom{n-m}{k-m}} =$	Frees the running index k in a nested binomial product.
$\frac{\binom{n}{k} \binom{n-k}{m}}{\binom{n}{m} \binom{n-m}{k}} =$	Shows that disjoint committees can be chosen in either order. Proves the trinomial theorem.
Iterated $\prod \binom{a_{i-1}}{a_i}$	Counts chains in the Boolean lattice by independent shell-assignment.

Unifying picture

All three identities are different ways of saying the same thing: the multinomial coefficient $\frac{n!}{a_0! a_1! \cdots a_r!}$ counts a single object, and any grouping of the a_i leads to a valid factorisation.

Pattern Recognition Checklist

When you see a sum or product involving binomial coefficients, scan for these signatures.

- A running index appears in **two** binomial coefficients, one nested in the other $\binom{n}{k}\binom{k}{m}$: apply Identity 1 to free the index.
- A **polynomial weight** $k, k^2, k(k-1)$: use the $m = 1$ or $m = 2$ absorption. Rewrite k^2 as $k(k-1) + k$ first.
- A weight of $\frac{1}{k+1}$: use absorption to shift $(n, k) \rightarrow (n+1, k+1)$.
- An **alternating sign** $(-1)^{k-m}$: expect a collapse to $(1-1)^{n-m} = \delta_{n,m}$.
- An **integral of a polynomial** on $[0, 1]$: expect the Beta function $B(m+1, n-m+1) = \frac{m!(n-m)!}{(n+1)!}$.
- **Two independent indices** in nested binomials $\binom{n}{k}\binom{n-k}{m}$: apply Identity 2 for a multinomial rewrite.
- A **chain of nested subsets**: apply Identity 3; often the answer is $(r+1)^n$ or a similar exponential.

Practice Problems (1/2)

Work these before next class. Send me the solutions to check.

Problem A

Evaluate $\sum_{k=0}^n \binom{n}{k} \binom{k}{\lfloor n/2 \rfloor}$.

Problem B

Find a closed form for $\sum_{k=0}^n k(k-1) \cdots (k-r+1) \binom{n}{k}$, where $0 \leq r \leq n$.

Problem C

Evaluate $\sum_{k=0}^n \frac{k}{k+1} \binom{n}{k}$.

Practice Problems (2/2)

Problem D

Show that for any prime p and $0 < k < p$, p divides $\binom{p}{k}$. Use Theorem 2.

Problem E

Count the number of chains $\emptyset \subsetneq A_1 \subsetneq A_2 \subsetneq \cdots \subsetneq A_r \subsetneq \{1, \dots, n\}$ with strict inclusions.

Problem F

Using the identity $\binom{n}{k} \binom{k}{m} = \binom{n}{m} \binom{n-m}{k-m}$, give a new proof of Vandermonde's identity $\sum_k \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}$.

Summary

- One identity: $\binom{n}{k}\binom{k}{m} = \binom{n}{m}\binom{n-m}{k-m}$.
- One move: free the running index so it appears in only one binomial.
- Five families of applications:
 - Binomial sums in closed form.
 - Polynomial-weighted sums via falling-factorial absorption.
 - Orthogonality and Möbius inversion.
 - Divisibility and the Frobenius map.
 - Multinomial and lattice-chain counting.
- The identity is elementary to state, elementary to prove, and extraordinarily productive. It is a good example of how a single combinatorial fact, used carefully, replaces a great deal of calculus and generating-function machinery.

End of class. Work the practice problems.